Business Requirements Document (BRD)

Multi-Module Enterprise Resource Planning (ERP) System

Buch Athletics, Pakistan

Methodology reference: Kendall & Kendall, *Systems Engineering Analysis and Design*, 11th Ed.

Abstract

This document presents the business requirements for an integrated Enterprise Resource Planning (ERP) system designed for Buch Athletics, a health club management business operating membership, scheduling, facility, and staff management functions through largely manual and fragmented processes. The system's scope is decomposed into four sequential milestones: Core Infrastructure & Revenue Operations; Operational Efficiency & Staff Management; Member Engagement & Experience; Ancillary Revenue & Analytics. Each milestone is dependent on the stability of the prior phase.

The proposed architecture applies systems analysis principles to structure the ERP as a three-tier, service-oriented system spanning financial management, member/CRM management, human resources, facility booking, member engagement, and business intelligence, with select commoditized functionality (HR/payroll) integrated via an open-source module to reduce build time while custom development is reserved for the business's differentiating workflows (Poba-Nzaou & Raymond, 2013).

1. Problem Statement

Buch Athletics currently manages membership, billing, scheduling, and facility operations through manual or fragmented processes; spreadsheet-based member records, paper-based scheduling, and manual billing reconciliation. This creates three compounding problems: (1) revenue leakage from missed renewals and manual billing errors; (2) data fragmentation, with no single source of truth for member, financial, or operational data, preventing consolidated reporting; (3) limited operational visibility, with no systematic way to track facility utilization, staff efficiency, or member retention. No integrated audit trail currently exists to support financial compliance or expense-fraud detection.

2. Proposed Solution

The proposed system is a modular ERP built around a three-tier architecture with a service-oriented integration layer, sequenced for delivery across four milestones rather than deployed as a single monolithic release.

Milestone 1: Core Infrastructure & Revenue Operations

Automated membership billing, complaint payment processing, basic P&L reporting, member registration with tiered plans, and biometric check-in. This milestone is prioritized first because it directly stops revenue leakage and establishes the authoritative member data source that every later module depends on. It also prioritizes digital membership records and efficient member check-in to improve customer experience.

Milestone 2: Operational Efficiency & Staff Management

HRM module for shift scheduling, payroll, biometric attendance sync, integrated automated expense tracking with fraud detection (receipt hashing, OCR validation, policy enforcement, immutable audit trail), a concurrency-safe facility booking engine (Redis-cached availability with async queue-based confirmation and waitlist auto-promotion), and class/event scheduling with automated reminders.

Milestone 3: Member Engagement & Experience

A member-facing mobile application to facilitate bookings, schedules, payment history, OAuth/2FA authentication, a complaint/feedback module with sentiment analysis, fitness tracking, and CRM-driven retention campaigns including inactivity detection, renewal reminders, referral tracking.

Milestone 4: Ancillary Revenue & Analytics

Café/retail point-of-sale, a loyalty points program, an online store, and a business intelligence dashboard reporting revenue by source, retention, utilization. This is deliberately sequenced last because meaningful analytics require 6+ months of stable operational data.

All modules communicate through a microservices application layer via REST APIs behind an API gateway, backed by PostgreSQL for transactional data, MongoDB for unstructured data (feedback, fitness logs), and Redis for caching and asynchronous task queuing (Reitsma & Per, 2018).

3. Project Scope

In scope:

- Financial management: automated billing, payment gateway integration, revenue tracking, P&L reporting.
- Member management and CRM: registration, tiered plans, digital profiles, biometric check-in, retention campaigns.
- Human resources management: staff scheduling, payroll, attendance tracking.
- Expense management: automated categorization, GL coding, fraud detection, and immutable audit logging.
- Facility and class/event booking with concurrency handling and waitlist/refund workflows.
- Member-facing mobile application and self-service portal.
- Complaint/feedback management (CFM) with sentiment analysis.
- Optional fitness tracking with wearable API integration (deprioritized).
- Point-of-sale, loyalty program, and online store (deprioritized).

- Business intelligence and analytics dashboard.

Out of scope:

- Multi-location or franchise management (single-facility deployment).
- Third-party gym-equipment IoT telemetry beyond biometric check-in hardware.
- Full e-commerce fulfillment/logistics beyond order-status tracking.
- Physical training staff operations and KPI module.

4. Business Objectives

1. Eliminate revenue leakage from manual billing and membership tracking, targeting a billing error rate below 1% (versus an estimated ~10% under manual processing).
2. Establish a single, authoritative source of truth for member, financial, and operational data across all modules.
3. Reduce administrative overhead through automation of scheduling, payroll, and expense workflows, targeting reduction in staff time spent on admin tasks.
4. Improve member retention through self-service tools and data-driven engagement.
5. Enable data-driven decision-making through consolidated business intelligence once sufficient operational data exists.

6. Functional Requirements

Milestone 1: Core Infrastructure & Revenue Operations

Priority: Generate immediate ROI and establish the data foundation.

- FR1.1: System shall process membership billing automatically on a recurring schedule (monthly/quarterly/yearly).
- FR1.2: System shall integrate with a PCI-DSS compliant payment gateway (IBFT/JazzCash/EasyPaisa) for transaction processing.
- FR1.3: System shall generate basic profit-and-loss (P&L) reports from transactional data.

- FR1.4: System shall support member registration with tiered plan selection and digital profile creation, including photo ID capture.
- FR1.5: System shall maintain a centralized member database (basic CRM) including contact info, membership history, and emergency contacts.
- FR1.6: System shall support biometric (fingerprint/facial recognition) check-in for attendance tracking.
- **Output:** Stops revenue leakage from manual billing; establishes a single source of truth for member data.

Milestone 2: Operational Efficiency & Staff Management

Priority: Automate repetitive tasks and employee oversight.

- FR2.1: System shall support shift scheduling for trainers and front-desk staff.
- FR2.2: System shall process payroll, integrated with biometric attendance data and secure banking gateways compliant with Pakistan FBR standards.
- FR2.3: System shall identify and automate recurring expenses, pushing approved expenses to the finance module with correct GL coding.
- FR2.4: System shall verify receipt authenticity via hash comparison (MD5/SHA-256) to flag duplicate or mismatched submissions.
- FR2.5: System shall apply OCR-based automated categorization and validation to submitted expense receipts.
- FR2.6: System shall enforce a configurable expense policy schema, including document retention requirements and policy-override logging for admin approval.

- FR2.7: System shall maintain an immutable audit trail with fraud-detection logic for all expense transactions.
- FR2.8: System shall support real-time facility booking (equipment, pool, studio) with a cached availability service to handle concurrent requests.
- FR2.9: System shall support waitlist auto-promotion and refund workflows for bookings.
- FR2.10: System shall support class/event scheduling with trainer assignment and automated SMS/email reminders.
- FR2.11: System shall enforce biometric/RFID-based facility access authorization.

Output: Reduces staff overhead costs, improves scheduling reliability, provides operational visibility.

Milestone 3: Member Engagement & Experience

Priority: Retention and self-service automation.

- FR3.1: System shall provide a member-facing mobile application supporting bookings, class schedules, attendance history, and payment receipts.
- FR3.2: System shall send push notifications for booking confirmations, reminders, and promotions.
- FR3.3: System shall support OAuth-based delegated authentication and two-factor authentication for member app access.
- FR3.4: System shall provide a complaint/feedback module (CFM) with ticket routing, response SLAs, and sentiment analysis dashboarding.

- FR3.5: System shall support fitness tracking including body measurement logging, workout plan assignment, progress dashboards, and optional wearable API integration.
- FR3.6: System shall trigger automated renewal reminders at 30/15/7 days before membership expiry.
- FR3.7: System shall support referral program tracking and lead-nurturing workflows.
- FR3.8: System shall identify inactive members (no bookings in 30 days) for targeted retention campaigns.
- FR3.9: System shall optionally calculate a predictive member health/cancellation-risk score based on engagement signals.
- FR3.10: System shall recommend classes based on historical attendance patterns and support facility-booking upsell prompts for frequent attendees.

Output: Builds on stable operations; self-service reduces front-desk burden; engagement data informs retention strategy.

Milestone 4: Ancillary Revenue & Analytics

Priority: Maximize per-member revenue and business intelligence.

- FR4.1: System shall support point-of-sale transactions for café/merchandise, linked to member accounts, with inventory management and low-stock alerts.
- FR4.2: System shall support a loyalty points program across membership referrals and café/retail purchases, with redemption tracking.
- FR4.3: System shall support an online store module for supplements/merchandise with payment gateway integration and order-fulfillment tracking.

- FR4.4: System shall provide a business intelligence dashboard reporting revenue by source, member retention rate, peak usage times, trainer utilization, and churn prediction.

  (For detailed process workflows see section 13.)

- **Non-Functional Requirements**

- NFR1 (Performance): Member check-in shall complete in under 15 seconds (versus 2–3 minutes under the prior manual process).
- NFR2 (Reliability): Billing error rate shall remain below 1%.
- NFR3 (Scalability): The booking system shall handle concurrent requests via cached availability and asynchronous event-queue updates without double-booking.
- NFR4 (Security): All payment processing shall use PCI-DSS compliant gateways; no raw card data shall be stored locally.
- NFR5 (Auditability): All financial transactions and policy overrides shall be logged in an immutable, timestamped audit trail.
- NFR6 (Availability): Milestone 1 financial and member-management modules shall maintain high uptime, and contingency palns.
- NFR7 (Data Integrity): Legacy data migration shall be validated via a test migration on a 10% data sample before full cutover. (milestone 1)
- NFR8 (Interoperability): All modules shall communicate via documented REST APIs through a central API gateway, enabling independent module updates.

Table 1

Stakeholders and RACI Overview

Stakeholder	Interest	Role in Rollout
Executive/Ownership	ROI, cash flow visibility, revenue growth	Accountable for final sign-off per milestone

Front-desk & Operations Staff	Ease of use, reduced manual admin work	Responsible for daily system operation
Finance/Accounting	Accurate billing, GL coding, audit trail	Responsible for financial data validation
HR/Staff Management	Scheduling accuracy, payroll automation	Responsible for HRM module validation
Trainers/Instructors	Class scheduling, booking visibility	Consulted for workflow input
Members (end users)	Self-service booking, transparent billing	Informed by primary end users, Milestone 3+
IT/Development Team	Maintainability, integration stability, security	Responsible for build and deployment

7. System Architecture

Tier 1: Presentation Layer

Web dashboard (React/Next.js) for staff/admin use; mobile application (React Native) for member self-service; optional tablet kiosks for facility-entrance check-in.

Tier 2: Application Layer (Service-Oriented)

Microservices architecture, one service per functional module (Member, Billing, Booking, HRM, POS), communicating via REST APIs behind a central API gateway for authentication, rate limiting, and routing (Slivka, 2024). A Redis-backed message queue handles asynchronous tasks like email/SMS notifications and payment processing callbacks

Tier 3: Data Layer

PostgreSQL for transactional data (members, bookings, payments), chosen for ACID compliance and JSON support; MongoDB for unstructured data (feedback, fitness logs); Redis cache for session management and real-time booking-availability queries.

Tier 4: Integration Layer

Third-party integrations for payment gateways (JazzCash/EasyPaisa), SMS (Twilio), biometric hardware, and ~~optional wearable fitness-tracker APIs~~. The CRM, HRM, booking engine, and expense modules are custom-built in-house, since these functions are directly tied to the business's retention and risk-control objectives and are not well served by generic tooling (Hustad & Stensholt, 2023)..

Table 2

Recommended Development Stack

Component	Technology	Rationale
Backend	Node.js + Express, or Python + Django	Fast API development, large ecosystem, ERP-friendly
Frontend	React + Next.js	Component reusability; SEO benefit for the online store
Mobile	React Native	Single codebase for iOS/Android, shares logic with web frontend
Database	PostgreSQL (primary) + Redis (cache)	ACID compliance, JSON support, horizontal scalability
Unstructured data store	MongoDB	Flexible schema for feedback and fitness-log data
Hosting	On-premises	Based on proprietary server availability
CI/CD	GitHub Actions + Docker	Automated testing/deployment, containerized consistency across environments

8. Data Model Overview

Core entities: Member, Membership Plan, Transaction, Attendance, Employee, Shift, Payroll, Expense, Resource, Booking, Event, Feedback, Fitness Log, Loyalty Account, Inventory Item, POS Transaction, and Audit Log. 17 entities in total, related primarily through the Member entity (billing, bookings, feedback, fitness logs, loyalty) and the Employee entity (shifts, payroll, events, expenses). A full field-level data dictionary and entity-relationship diagram accompany this document as a supporting artifact.

(Find detailed Data Dictionary in section 14.)

9. Security, Compliance, and Fraud Controls

- Payment processing restricted to PCI-DSS compliant gateways; no raw card data stored locally.
- Expense fraud prevention via receipt hashing (duplicate detection), OCR-based validation, and frequency analysis to detect expense-splitting patterns.
- Immutable audit trail for all financial transactions and policy overrides, with anomaly flagging for admin review.
- OAuth-based delegated access and two-factor authentication for the member-facing application.
- Daily automated backups to separate cloud storage, with monthly restore testing.

10. Feasibility Analysis

- **Technical feasibility:** The three-tier architecture and REST-based service integration are established patterns for multi-module business systems.
- **Economic feasibility:** Custom development for core modules is development intensive and extends timeline for development. This increases risks for scope-creep, integration failures, hardware failures and requires high training requirement for staff and management. However, management has prioritized in-house development and hosting (Lee et al., 2024).
- **Operational feasibility:** The phased milestone structure allows staff to adopt the system incrementally. Milestone 1's revenue-critical functions stabilize before staff must learn

HRM, booking, and engagement tools in later phases supported by a three-phase training plan and department-level support model.

- **Risk feasibility:** Sequencing analytics (Milestone 4) after 6+ months of operational data collection avoids building business intelligence on an unstable or insufficient data foundation; parallel analog/digital operation during Milestone 1 directly mitigates data-migration risk.

11. Implementation Roadmap

Milestone	Focus	Key Deliverables
1	Core Infrastructure & Revenue Operations	Billing automation, member core, basic CRM, biometric entry
2	Operational Efficiency & Staff Management	HRM expense automation, booking engine, event management
3	Member Engagement & Experience	Mobile app, CFM, fitness tracking, advanced CRM campaigns
4	Ancillary Revenue & Analytics	POS, loyalty, online store, BI dashboard

Change management: Three-phase staff training (admin overview, hands-on modules, ongoing support), with department "super users" as first-line support. Parallel analog/digital operation for 1–2 months during Milestone 1 to mitigate resistance and catch data migration errors.

Data migration: Legacy spreadsheet/paper records exported to CSV, deduplicated, and validated via a 10% test migration before full cutover.

12. Assumptions and Constraints

- Single-facility deployment is assumed; multi-location/franchise scaling is out of scope and would require re-architecting the data layer for multi-tenancy.
- Hosting is assumed on-premises based on proprietary server availability; this should be revisited if infrastructure constraints change.

- Wearable fitness-tracker integration and AI-powered churn-risk scoring (Milestone 3) are optional and should be validated against member demand and data availability before development investment. (out-of-scope)
- Payment processing assumes access to PCI-DSS compliant gateways operable in Pakistan (e.g., IBFT, JazzCash, EasyPaisa).

Table 3

Risk Register

Risk	Likelihood	Impact	Mitigation
Data migration errors from legacy spreadsheets	Medium	High	10% test-sample migration before full cutover; parallel operation for 1–2 months
Staff resistance to new system	Medium	Medium	Phased training, departmental "super users," parallel analog operation
Payment gateway downtime	Low	High	Retry logic on billing job; reconciliation against bank settlement reports
Booking race conditions under peak load	Medium	Medium	Redis-cached availability + async queue with waitlist auto-promotion
Expense fraud (duplicate/split receipts)	Medium	Medium	Hashing, OCR validation, frequency analysis, immutable audit trail
Analytics module launched before sufficient data	Low (if roadmap followed)	Medium	Milestone 4 deliberately sequenced after 6+ months of operational data

Scope creep into multi-location support	Medium	High	Explicitly out-of-scope; would require data-layer re-architecture
Vendor/API dependency (payment gateway, SMS, biometric SDK)	Low	Medium	Abstracted via API Gateway to ease future vendor swaps

13. Detailed Process Workflows

Membership Billing Cycle (Milestone 1)

1. System identifies members due for renewal based on billing_cycle_days from Membership Plan.
2. Automated billing job triggers payment request via the PCI-DSS compliant gateway.
3. On success: transaction recorded, member status remains active, receipt sent via SMS/email.
4. On failure: retry logic triggers (configurable attempts), member flagged for front-desk follow-up if retries exhausted.
5. Finance module reconciles gateway reference against bank settlement reports.

Facility Booking with Concurrency Control (Milestone 2)

1. Member/staff requests a booking for a resource and time slot.
2. System checks Redis-cached availability for that resource/slot.
3. If available: booking tentatively reserved, event pushed to async queue for confirmation write to PostgreSQL.
4. If slot fills before confirmation completes (race condition): requester is placed on waitlist automatically.
5. On cancellation: waitlisted member is auto-promoted; refund workflow triggers if payment was already collected.

Expense Submission and Fraud Check (Milestone 2)

1. Employee submits expense with receipt image.

2. System computes SHA-256 hash of receipt image; checks against existing hashes for duplicates.
3. OCR extracts vendor, amount, date, tax number; auto-categorizes expense.
4. System cross-checks extracted data against policy schema (spend limits, allowed categories).
5. If flagged (duplicate, policy violation, or frequency-analysis anomaly suggesting expense-splitting): routed to admin for manual review with override capability (logged).
6. If clean: auto-approved, pushed to finance module with GL coding, and recorded in the immutable audit trail.

Member Retention Campaign Trigger (Milestone 3)

1. Nightly batch job scans booking/attendance history.
2. Members with no bookings in 30 days are flagged as "inactive."
3. CRM triggers a retention campaign (email/push notification) with a configurable offer or check-in message.
4. Engagement response (booking made, app opened) is logged and feeds back into the optional cancellation-risk scoring model.

Security, Compliance, and Fraud Controls

- **Payment security**: PCI-DSS compliant gateways only; no raw card data stored locally.
- **Expense fraud prevention**: receipt hashing for duplicate detection; OCR-based validation; metadata and frequency analysis to detect expense-splitting; tax-number verification for receipt authenticity.
- **Audit trail**: immutable logging for all financial transactions and policy overrides, with fraud-detection algorithms flagging anomalies for admin review.
- **Authentication**: OAuth-based delegated access and two-factor authentication for the member-facing application.
- **Data retention**: expense documentation retained per FBR-aligned compliance policy.

- **Backup and disaster recovery**: daily automated backups to separate cloud storage (AWS S3); restore procedures tested monthly.

14. Detailed Data Dictionary

Member

Field	Type	Notes
member_id	UUID (PK)	Primary identifier
full_name	String	
photo_id_url	String	Stored reference, not raw binary in DB
membership_tier_id	FK → Membership Plan	
join_date	Date	
emergency_contact	String	
status	Enum (active/expired/suspended)	

Membership Plan

Field	Type	Notes
plan_id	UUID (PK)	
tier_name	String	e.g., Monthly, Quarterly, Yearly
billing_cycle_days	Integer	
price	Decimal	
renewal_terms	Text	

Billing/Transaction

Field	Type	Notes
transaction_id	UUID (PK)	
member_id	FK → Member	

amount	Decimal	
payment_method	Enum	IBFT/JazzCash/EasyPaisa/Card
gateway_reference	String	External reference for reconciliation
gl_code	String	For finance module integration
status	Enum (success/failed/refunded)	
timestamp	DateTime	

Attendance/Check-in

Field	Type	Notes
checkin_id	UUID (PK)	
member_id	FK → Member	
timestamp	DateTime	
biometric_method	Enum (fingerprint/facial)	
facility_zone	String	

Staff/Employee

Field	Type	Notes
employee_id	UUID (PK)	
role	Enum (trainer/front-desk/admin)	
shift_schedule_ref	FK → Shift table (Frappe HR integration)	
payroll_ref	FK → Payroll record	

Expense

Field	Type	Notes
expense_id	UUID (PK)	

submitted_by	FK → Employee	
receipt_hash	String (SHA-256)	Duplicate detection
category	Enum	
gl_code	String	
approval_status	Enum (pending/approved/rejected/overridden)	
audit_log_ref	FK → Audit Log	

Facility/Resource

Field	Type	Notes
resource_id	UUID (PK)	
type	Enum (equipment/pool/studio)	
capacity	Integer	
availability_cache_key	String	Redis key reference

Booking

Field	Type	Notes
booking_id	UUID (PK)	
member_id	FK → Member	
resource_id	FK → Resource	
time_slot	DateTime range	
status	Enum (confirmed/waitlisted/cancelled)	
refund_status	Enum (n/a/pending/completed)	

Class/Event

Field	Type	Notes
event_id	UUID	

trainer_id	FK → Employee	
schedule	DateTime	
capacity	Integer	
reminder_status	Enum (pending/sent)	

Feedback/Complaint

Field	Type	Notes
ticket_id	UUID (PK)	
member_id	FK → Member	
category	String	
sentiment_score	Decimal	From sentiment analysis pipeline
sla_status	Enum (on-track/breached)	
routing_history	JSON	

Fitness Log

Field	Type	Notes
log_id	UUID (PK)	
member_id	FK → Member	
measurement_type	Enum (weight/BMI/other)	
value	Decimal	
date	Date	
source	Enum (manual/wearable)	

Loyalty Account

Field	Type	Notes

member_id	FK → Member (PK)	
points_balance	Integer	
redemption_history	JSON	

POS Transaction / Inventory Item

Field	Type	Notes
pos_transaction_id	UUID (PK)	
member_id	FK → Member (nullable)	
items	JSON array of item_id + qty	
inventory_item_id	FK → Inventory Item	
stock_level	Integer	
reorder_threshold	Integer	

Conclusion

This document has outlined the business case, requirements, and system design for a multi-module ERP serving Buch Athletics' membership, financial, staff, and facility operations. The proposed four-milestone sequencing — Core Infrastructure & Revenue Operations, Operational Efficiency & Staff Management, Member Engagement & Experience, and Ancillary Revenue & Analytics — directly addresses the three compounding problems identified in Section 1: revenue leakage from manual billing, fragmented member data with no single source of truth, and limited operational visibility into facility and staff performance.

Sequencing analytics last, after the core transactional modules have accumulated 6+ months of stable operational data, is a deliberate risk-mitigation decision rather than a deprioritization of business intelligence. Attempting to build meaningful dashboards on an unstable or incomplete data foundation is a common and avoidable failure mode in ERP implementations, and this roadmap is structured specifically to avoid it (Uppstrom et al., 2015).

Successful delivery depends on three factors outside the system's technical design: disciplined data migration, sustained staff adoption through the three-phase training and "super user" support model, and continued stakeholder alignment as each milestone's output informs the scope of the next. Subject to these conditions, the proposed system is expected to deliver measurable, trackable impact within the first two milestones, with the full revenue and analytics is realized by the end of Milestone 4.

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